EDGE DEVICE MEASUREMENT REPORT

Electronic Nose Coffee Classification System

Platform: NVIDIA Jetson Orin Nano

Model: Convolutional Neural Network (1D-CNN)

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MEASUREMENT SUMMARY

Single-Sample Inference Latency	Avg Total Power (Training)	Training Time
0.954360 ms	**9.81 W**	**20.20 min**

MODEL CONFIGURATION

Model Type	1D Convolutional Neural Network (1D-CNN)
Epochs	50
Input Shape	8 sensors x 300 timesteps
Architecture	1D-CNN + Backpropagation (SGD, Cross Entropy)
Single-Sample Inference Latency	0.954360 ms
Model Size	923.44 KB
Number of Parameters	182978
Memory Footprint	5649.00 MB
Power Measurement	9.81 W

POWER, THERMAL & RESOURCE MEASUREMENTS DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	9.81 W	**Avg CPU Temp**	54.5 C
Peak Total Power	11.31 W	**Peak CPU Temp**	57.1 C
Avg CPU+GPU Power	4.55 W	**Avg GPU Temp**	53.5 C
Peak CPU+GPU Power	5.89 W	**Peak GPU Temp**	55.9 C
Avg RAM Usage	4234 MB	**Avg GPU Util**	52.8 %
Peak RAM Usage	5649 MB	**Peak GPU Util**	99.0 %
Energy Used	11472.7 J (3.1869 Wh)	**Training Duration**	20.20 min

GAS SENSORS USED (8 Sensors)

TGS2600	**MQ135**	**MQ3**	**MQ6**
MQ7	**TGS2602**	**TGS2611**	**TGS2620**